

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihal Produk:

Product Description:

Cat No. :

Synonyms

Molecular Formula

**AQUALINE® COMPLETE 2**
**Aqualine™ Complete 2**

K/1950/08, K/1950/15, K/1950/17

Karl Fischer Reagent

Solution

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

 Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888

 CHEMTREC Malaysia **1-800-815-308** (Malay)

 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                     |
|------------------------------------------------------|---------------------|
| Skin Corrosion/Irritation                            | Category 2 (H315)   |
| Serious Eye Damage/Eye Irritation                    | Category 2 (H319)   |
| Reproductive Toxicity                                | Category 1B (H360D) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373)   |

**Label Elements**


Signal Word

Danger

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## Hazard Statements

H315 - Causes skin irritation  
H319 - Causes serious eye irritation  
H360D - May damage the unborn child  
H373 - May cause damage to organs through prolonged or repeated exposure

## Precautionary Statements

### Prevention

P201 - Obtain special instructions before use  
P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P202 - Do not handle until all safety precautions have been read and understood  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P280 - Wear eye protection/ face protection

### Response

P308 + P313 - IF exposed or concerned: Get medical advice/attention  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P332 + P313 - If skin irritation occurs: Get medical advice/attention  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 - Store in a well-ventilated place

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                                  | CAS No    | Weight % |
|--------------------------------------------|-----------|----------|
| Diethylene glycol monoethyl ether          | 111-90-0  | 75 - 90  |
| [(Imidazol-1-yl)sulfonyl]oxyethoxydiglycol | NA        | 5 - 15   |
| Iodine                                     | 7553-56-2 | < 5      |
| 1-Imidazole                                | 288-32-4  | 2.5 - 5  |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

#### Inhalation

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device.

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Immediate medical attention is required.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable. Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

Thermal decomposition can lead to release of irritating gases and vapors. Containers may explode when heated. Keep product and empty container away from heat and sources of ignition.

**Hazardous Combustion Products**

Carbon monoxide (CO), Sulfur oxides, Nitrogen oxides (NO<sub>x</sub>), Carbon dioxide (CO<sub>2</sub>).

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Use personal protective equipment as required. Ensure adequate ventilation. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental precautions**

Do not flush into surface water or sanitary sewer system.

**Methods and Material for Containment and Cleaning Up**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

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## Precautions for Safe Handling

Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

## Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Protect from direct sunlight. Should crystals form in a peroxidizable liquid, peroxidation may have occurred and the product should be considered extremely dangerous. In this instance, the container should only be opened remotely by professionals.

## Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component | Malaysia | ACGIH TLV              | OSHA PEL                                                                                                                 |
|-----------|----------|------------------------|--------------------------------------------------------------------------------------------------------------------------|
| Iodine    |          | TWA: 0.001 ppm<br>Skin | Ceiling: 0.1 ppm<br>Ceiling: 1 mg/m <sup>3</sup><br>(Vacated) Ceiling: 0.1 ppm<br>(Vacated) Ceiling: 1 mg/m <sup>3</sup> |

| Component                         | European Union | The United Kingdom                                         | Germany                                                                                                                                                                                                                                             |
|-----------------------------------|----------------|------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Diethylene glycol monoethyl ether |                |                                                            | TWA: 6 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 35 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 50 mg/m <sup>3</sup> (8 Stunden). MAK can occur as vapor and aerosol at the same time<br>Höhepunkt: 100 mg/m <sup>3</sup> |
| Iodine                            |                | STEL: 0.1 ppm 15 min<br>STEL: 1.1 mg/m <sup>3</sup> 15 min | Haut                                                                                                                                                                                                                                                |

### Exposure Controls

#### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Goggles

#### Hand Protection

Protective gloves

#### Skin and body protection

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use

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## Recommended Filter type:

appropriate certified respirators  
Organic gases and vapours filter Type A Brown conforming to EN14387  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
When RPE is used a face piece Fit Test should be conducted

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

Prevent product from entering drains Do not allow material to contaminate ground water system

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

**Appearance** Brown  
**Physical State** Liquid  
**Odor** Alcohol-like  
**Odor Threshold** No data available  
**pH** No information available

**Melting Point/Range** No data available  
**Softening Point** No data available  
**Boiling Point/Range** No information available  
**Flash Point** No information available **Method -** No information available

**Evaporation Rate** No data available  
**Flammability (solid,gas)** Not applicable  
**Explosion Limits** No data available  
Liquid

**Vapor Pressure** No data available  
**Vapor Density** No data available (Air = 1.0)  
**Specific Gravity / Density** 1.09  
**Bulk Density** Not applicable  
**Water Solubility** Miscible  
**Solubility in other solvents** No information available  
Liquid

### Partition Coefficient (n-octanol/water)

| Component                         | log Pow |
|-----------------------------------|---------|
| Diethylene glycol monoethyl ether | -0.8    |
| Iodine                            | 2.49    |
| 1-Imidazole                       | -0.02   |

**Autoignition Temperature** No data available  
**Decomposition Temperature** No data available  
**Viscosity** No data available  
**Explosive Properties** No information available  
**Oxidizing Properties** No information available

**Molecular Formula** Solution

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## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions. May form explosive peroxides.

### Possibility of Hazardous Reactions

#### **Hazardous Polymerization Hazardous Reactions**

Hazardous polymerization does not occur.  
None under normal processing.

### Conditions to Avoid

Incompatible products. Heat, flames and sparks. Extremes of temperature and direct sunlight.

### Incompatible Materials

Strong oxidizing agents. Reducing Agent. Strong acids. Bases. Acid anhydrides. Acid chlorides. Metals.

### Hazardous Decomposition Products

Carbon monoxide (CO). Sulfur oxides. Nitrogen oxides (NOx). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### **Product Information**

##### **(a) acute toxicity;**

**Oral**

Based on available data, the classification criteria are not met

**Dermal**

Based on available data, the classification criteria are not met

**Inhalation**

Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component                         | LD50 Oral          | LD50 Dermal                                                     | LC50 Inhalation                           |
|-----------------------------------|--------------------|-----------------------------------------------------------------|-------------------------------------------|
| Diethylene glycol monoethyl ether | 6031 mg/kg ( Rat ) | 9143 mg/kg (Rabbit)<br>4200 µL/kg ( Rabbit )<br>6 mL/kg ( Rat ) | LC50 > 5240 mg/m <sup>3</sup> ( Rat ) 4 h |
| Iodine                            | 315 mg/kg ( Rat )  | 1425 mg/kg ( Rabbit )                                           | 4.588 mg/L 4h ( Rat )                     |
| 1-Imidazole                       | 970 mg/kg (Rat)    | -                                                               | -                                         |

**(b) skin corrosion/irritation;** Category 2

**(c) serious eye damage/irritation;** Category 2

**(d) respiratory or skin sensitization;**

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|                                                                  |                                                                                                                                     |                     |                     |
|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|---------------------|---------------------|
| <b>Respiratory</b>                                               | No data available                                                                                                                   |                     |                     |
| <b>Skin</b>                                                      | No data available                                                                                                                   |                     |                     |
| <b>Component</b>                                                 | <b>Test method</b>                                                                                                                  | <b>Test species</b> | <b>Study result</b> |
| Iodine<br>7553-56-2 ( < 5 )                                      | OECD Test Guideline 429<br>Local Lymph Node Assay                                                                                   | mouse               | non-sensitising     |
| <b>(e) germ cell mutagenicity;</b>                               | No data available                                                                                                                   |                     |                     |
| <b>(f) carcinogenicity;</b>                                      | No data available                                                                                                                   |                     |                     |
|                                                                  | There are no known carcinogenic chemicals in this product                                                                           |                     |                     |
| <b>(g) reproductive toxicity;</b><br><b>Reproductive Effects</b> | Category 1B<br>Experiments have shown reproductive toxicity effects on laboratory animals.                                          |                     |                     |
| <b>(h) STOT-single exposure;</b>                                 | No data available                                                                                                                   |                     |                     |
| <b>(i) STOT-repeated exposure;</b>                               | Category 2                                                                                                                          |                     |                     |
| <b>Target Organs</b>                                             | Reproductive System.                                                                                                                |                     |                     |
| <b>(j) aspiration hazard;</b>                                    | No data available                                                                                                                   |                     |                     |
| <b>Other Adverse Effects</b>                                     | The toxicological properties have not been fully investigated.                                                                      |                     |                     |
| <b>Symptoms / effects,both acute and delayed</b>                 | Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting.                                                |                     |                     |
| <b>Endocrine Disrupting Properties</b>                           | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors. |                     |                     |

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

Toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component                         | Freshwater Fish                                                                                                                                                                                                                                                  | Water Flea                                  | Freshwater Algae                                                        | Microtox                                                    |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------|
| Diethylene glycol monoethyl ether | LC50: 11600 - 16700 mg/L, 96h flow-through (Pimephales promelas)<br>LC50: 11400 - 15700 mg/L, 96h flow-through (Oncorhynchus mykiss)<br>LC50: 19100 - 23900 mg/L, 96h flow-through (Lepomis macrochirus)<br>LC50: = 10000 mg/L, 96h static (Lepomis macrochirus) | EC50: 3940 - 4670 mg/L, 48h (Daphnia magna) |                                                                         |                                                             |
| Iodine                            | LC50 = 1.67 mg/L 96h                                                                                                                                                                                                                                             | EC50 = 0.55 mg/L 48h                        | EC50 = 0.13 mg/L 72h                                                    | EC50 = 280 mg/L 3h                                          |
| 1-Imidazole                       |                                                                                                                                                                                                                                                                  | EC50: = 341.5 mg/L, 48h (Daphnia magna)     | EC50: = 82 mg/L, 96h (Desmodesmus subspicatus)<br>EC50: = 130 mg/L, 72h | = 1200 mg/L EC50 Pseudomonas putida 17 h<br>= 231 mg/L EC50 |

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|  |  |  |                           |                                   |
|--|--|--|---------------------------|-----------------------------------|
|  |  |  | (Desmodesmus subspicatus) | Photobacterium phosphoreum 30 min |
|--|--|--|---------------------------|-----------------------------------|

## Persistence and degradability

### **Persistence**

Persistence is unlikely.

### **Degradation in sewage treatment plant**

Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants.

## Bioaccumulative potential

Bioaccumulation is unlikely

| Component                         | log Pow | Bioconcentration factor (BCF) |
|-----------------------------------|---------|-------------------------------|
| Diethylene glycol monoethyl ether | -0.8    | No data available             |
| Iodine                            | 2.49    | No data available             |
| 1-Imidazole                       | -0.02   | No data available             |

## Mobility in soil

The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

## Endocrine Disruptor Information

This product does not contain any known or suspected endocrine disruptors

## Other adverse effects

No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

### **Waste from Residues/Unused Products**

Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

### **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

### **International Inventories**

X = listed



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| Component                         | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Diethylene glycol monoethyl ether | 203-919-7 | X    | X   | X     | X    | X    | X     | X    | KE-10467 |
| Iodine                            | 231-442-4 | X    | X   | X     | X    |      | X     | X    | KE-21023 |
| 1-Imidazole                       | 206-019-2 | X    | X   | X     | X    | X    | X     | X    | KE-20937 |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Revision Date**

23-Mar-2025

**Revision Summary**

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**